

EnergyWise: LLM-Assisted Smart Campus Energy Optimization & Microgrid Digital Twin
Project Architecture, Machine Learning Pipeline, MILP Formulation, and Complete Faculty Defense Guide
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Date: September 2026

1. Executive Summary & Problem Framing

1.1 The Real-World Bangladesh Crisis

In Bangladesh, power outages (load-shedding) are not an occasional emergency demand-and-response event; they are routine grid management. The national peak electricity demand routinely outstrips maximum generation capacity by **1,000 to 1,500 MW**. Power distribution companies—such as the Dhaka Power Distribution Company (DPDC) and Dhaka Electric Supply Company (DESCO)—enforce severe **Time-of-Use (TOU) tariffs**:

- **Off-Peak Tariff (11:00 PM – 5:00 PM / 23:00 – 17:00):** ~7.0 BDT / kWh
- **Peak Hours Tariff (5:00 PM – 11:00 PM / 17:00 – 23:00):** ~12.0 BDT / kWh (a **+71.4% price surge**)

Institutional campuses (universities, medical complexes, research centers) consume vast amounts of electrical and thermal energy across dozens of distinct facilities. Under traditional building management:

1. Campuses behave as **passive consumers**, finding out about feeder disconnects at the exact instant power drops.
2. High-inertia HVAC loops, chillers, and empty lecture halls continue drawing full power during peak tariff windows.
3. Onsite battery energy storage systems (BESS) and rooftop photovoltaic (PV) arrays are dispatched naively or left uncoordinated, failing to protect critical academic and research facilities when grid shedding occurs.

1.2 The Solution: A Cyber-Physical Digital Twin

EnergyWise is an autonomous, closed-loop Cyber-Physical System (CPS) and digital twin designed to monitor, forecast, optimize, and protect a 24-building university campus in real time. It unifies three computing paradigms:

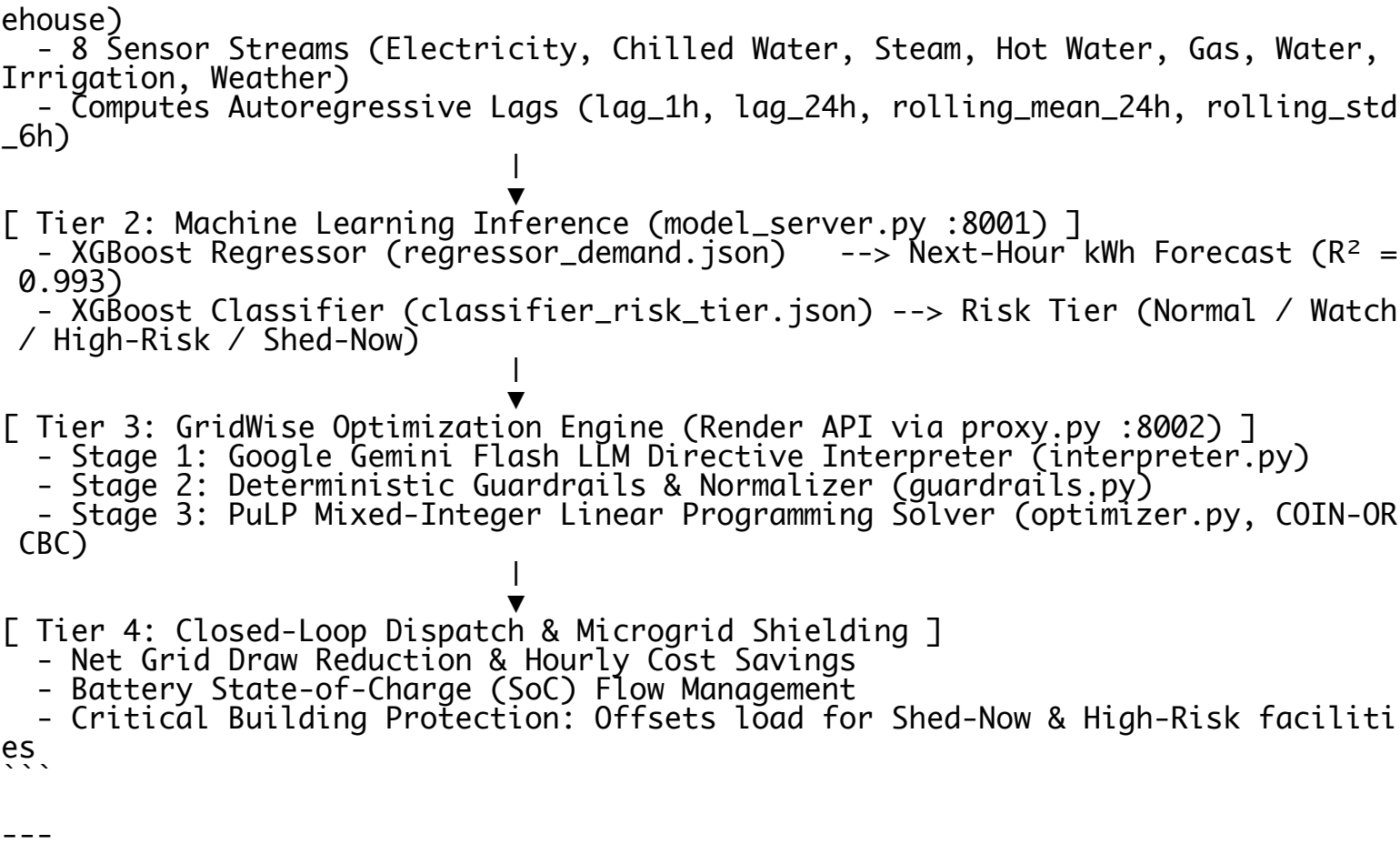
1. **Supervised Machine Learning (XGBoost):** Trained on 421,195 actual meter readings from real institutional buildings (Building Data Genome 2.0) to predict next-hour load and classify operational risk tiers.
2. **Generative Natural Language Processing (Google Gemini Flash LLM):** Ingests unstructured, conversational operator directives (e.g., **"Don't charge battery between 2-4 PM"** or **"DPDC warning: keep 50% battery in reserve"**) and converts them into structured machine constraints.
3. **Operations Research & Mathematical Optimization (PuLP MILP + COIN-OR CBC):** Formulates and solves a 24-hour Mixed-Integer Linear Program in under 20 milliseconds to guarantee the mathematically optimal grid/solar/battery dispatch that minimizes cost, obeys battery physics, and avoids peak tariffs.
4. **Closed-Loop Protection:** Local microgrid power (solar + battery discharge) is routed to protect at-risk campus buildings, actively preventing blackouts.

2. System Architecture

The project operates across a continuous four-tier pipeline:

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- [Tier 1: Real-Time Digital Twin Simulation (dashboard.html)]
 - 24 Building Twins (Office, Education, Lodging, Retail, Assembly, Public, War



3. The 24 Campus Buildings Directory

The campus digital twin models ****24 distinct building twins****, cycling across 7 standard building space types and 24 individual building identities.

3.1 Complete Building Inventory

Index	Building Identifier	Space Type	Architectural Role / Campus Function
1	Campus_office_Amy	Office	Central Faculty Offices & Academic Affairs
2	Campus_lodging_Bea	Lodging	Student Residential Hall / Dormitory Alpha
3	Campus_education_Cid	Education	Main Lecture Hall Complex & Classrooms
4	Campus_retail_Dan	Retail	Campus Cafeteria, Dining Hall & Bookstore
5	Campus_assembly_Eli	Assembly	Central University Auditorium & Event Arena
6	Campus_public_Fay	Public	Central University Library & Study Commons
7	Campus_warehouse_Gio	Warehouse	Central Facilities, IT Storage & Maintenance
8	Campus_office_Hana	Office	Vice-Chancellor & Administrative Secretariat
9	Campus_lodging_Ira	Lodging	International Scholar & Guest Residence
10	Campus_education_Joy	Education	Computer Science & Engineering Laboratories
11	Campus_retail_Kai	Retail	Student Center & Food Court Kiosks

12	`Campus_assembly_Lea`	Assembly	Multipurpose Gymnasium & Indoor Sports Facility	130 kWh	18,900 sq ft
13	`Campus_public_Moe`	Public	Student Admissions, Registrar & Career Wing	75 kWh	11,200 sq ft
14	`Campus_warehouse_Nia`	Warehouse	Groundskeeping Equipment & Utility Storage	40 kWh	5,500 sq ft
15	`Campus_office_Omar`	Office	Research Centers & Graduate Studies Building	105 kWh	16,300 sq ft
16	`Campus_lodging_Pia`	Lodging	Female Student Residential Dormitory Beta	115 kWh	18,700 sq ft
17	`Campus_education_Quin`	Education	Physics, Chemistry & Robotics Laboratories	120 kWh	16,800 sq ft
18	`Campus_retail_Rey`	Retail	Stationery, Convenience & Copy Center	50 kWh	4,900 sq ft
19	`Campus_assembly_Sam`	Assembly	Alumni Amphitheater & Performing Arts Studio	135 kWh	19,100 sq ft
20	`Campus_public_Tia`	Public	Campus Healthcare Center & Emergency Clinic	60 kWh	7,200 sq ft
21	`Campus_warehouse_Uma`	Warehouse	Central Chiller Plant & High-Voltage Switchgear	150 kWh	8,900 sq ft
22	`Campus_office_Vik`	Office	IT Infrastructure & High-Performance Datacenter	160 kWh	11,500 sq ft
23	`Campus_lodging_Wes`	Lodging	Male Student Residential Dormitory Gamma	110 kWh	17,900 sq ft
24	`Campus_education_Xia`	Education	Architecture, Civil Engineering & Design Studios	100 kWh	15,200 sq ft

3.2 Sensor Streams Tracked per Building

Each simulated building records 8 physical telemetry streams every hour:

1. `electricity`: Total electrical consumption (kWh)
2. `chilledwater`: Chilled water loop flow rate (tons/kWh equivalent)
3. `steam`: High-temperature steam flow for heating/sterilization
4. `hotwater`: Domestic hot water usage
5. `gas`: Natural gas flow for kitchens and boilers
6. `water`: General utility water meter (gallons/hr)
7. `irrigation`: Landscape irrigation flow (active predominantly at 05:00)
8. `hour`: Discrete hour index (\$0 \leq h \leq 23\$)

4. Digital Twin Simulation Engine (`dashboard.html`)

4.1 Diurnal Human Dynamics Modeling

Real campus demand follows non-linear human occupancy patterns. In `dashboard.html`, hourly electricity demand is calculated using a calibrated sinusoidal curve floored at 40%:

$$\text{dailyCurve}(h) = 0.4 + 0.6 \times \left(0.5 + 0.5 \times \sin\left(\frac{h - 6}{24} \times 2\pi\right)\right)$$

- **Physical Rationale:** Buildings never reach zero load overnight; baseline equipment (servers, emergency lighting, refrigeration, security routers) maintains continuous power draw.
- **Space Multipliers:** `office` and `education` facilities apply a **1.2x** weekday multiplier to reflect class schedules.
- **Stochastic Noise:** A randomized multiplier $\eta \sim \mathcal{U}(0.85, 1.15)$ simulates realistic occupancy volatility.

4.2 Environmental & Equipment Volatility Features

The digital twin simulates ambient microclimate conditions:

- **Air Temperature:** Sinusoidal cycle from 26°C to 32°C
- **Dew Point:** $20^{\circ}\text{C} \pm 1.5^{\circ}\text{C}$
- **Wind Speed:** $1.0 \text{ to } 5.0 \text{ m/s}$

Crucially, the simulation tracks a rolling history of the previous 30 hours to compute rolling standard deviations of HVAC telemetry:

$$\sigma_{\text{chilledwater, 6h}} = \sqrt{\frac{1}{6} \sum_{i=1}^6 (x_i - \bar{x})^2}$$

Why this matters: Sustained high chiller flow with zero variance ($\sigma \approx 0$) signals a stuck pneumatic actuator valve or an improperly tuned thermostat at loop—an equipment fault that simple static threshold rules cannot detect.

5. Machine Learning Models

5.1 Training Dataset: Building Data Genome 2.0 (BDG2)

Rather than training on synthetic approximations, our models were trained on the **Building Data Genome 2.0 (BDG2)** dataset (Lawrence Berkeley National Lab & National University of Singapore).

- **Scope:** 30 institutional buildings across an entire university campus.
- **Volume:** 421,195 actual hourly meter readings across 2016–2017.
- **Split Strategy:** Strict **chronological train/test split** (the earliest 80% of hours for training, the final 20% for testing). Random K-Fold splits were deliberately rejected because shuffling time-series data leaks future autoregressive information into past evaluations.

5.2 Model 1: XGBoost Regressor (`regressor\_demand.json`)

- **Objective:** Forecast building-level electricity demand ($y \in \mathbb{R}^+$) for the next operating hour.
- **Input Features (23 columns):** `hour`, `dayofweek`, `is\_weekend`, `month`, `lag\_1h`, `lag\_24h`, `rolling\_mean\_24h`, `airTemperature`, `dewTemperature`, `windSpeed`, `sqft`, plus meter lags and 6-hour rolling standard deviations for chilled water, steam, hot water, gas, water, and irrigation.
- **Empirical Test Metrics:**
 - Coefficient of Determination (R^2): 0.993
 - Mean Absolute Error (MAE): 5.97 kWh
 - Root Mean Squared Error (RMSE): 10.98 kWh
- **Feature Importance:** Autoregressive memory (`lag\_1h`, `lag\_24h`, `rolling\_mean\_24h`) accounts for ~98% of variance, confirming that immediate demand history is the single strongest physical predictor in high-frequency forecasting.
- **Production Guardrail:** Enforced via `max(0.0, demand)` clamp in `model_server.py` to prevent negative demand extrapolation on out-of-distribution inputs.

5.3 Model 2: XGBoost Classifier (`classifier\_risk\_tier.json`)

- **Objective:** Classify each building into an actionable operational risk tier based on historical percentile distributions:
 1. **Normal** ($< 75^{\text{th}}$ percentile of historical load): Baseline operations.
 2. **Watch** (75^{th} - 90^{th} percentile): Elevated consumption; monitoring required.
 3. **High-Risk** (90^{th} - 97^{th} percentile): Nearing branch breaker capacity.
 4. **Shed-Now** ($> 97^{\text{th}}$ percentile / Top 3%): Severe overload; immediate candidate for load shedding.
- **Empirical Performance & Class Balancing:** The classifier achieved an overall accuracy of 82.6%. However, because >75% of real building hours are **Normal**, a standard model achieves high overall accuracy while missing rare critical events (initial recall on **Shed-Now** was 0.20–0.23). We addressed this in `train_models.py` by applying inverse-frequency sample weighting:
$$w_c = \frac{N_{\text{total}}}{K \times N_c}$$
This heavily penalizes the model for failing to predict rare **Shed-Now** emergencies.

6. GridWise: The Mathematical Optimization & AI Engine

6.1 The 3-Stage Optimization Pipeline

GridWise separates natural language reasoning from rigorous numerical optimization:

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[Operator Note] $\rightarrow$ [Stage 1: Gemini Flash LLM] $\rightarrow$ [Stage 2: Guardrails]
 $\rightarrow$ [Stage 3: PuLP MILP Solver]

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1. **Stage 1 – LLM Directive Interpreter** (`interpreter.py`):** Translates free-form operator text into one of five structured directives: `solar_reduction`, `no_charge_window`, `no_discharge_window`, `minimum_battery_reserve`, or `no_op`.
2. **Stage 2 – Deterministic Guardrail Layer** (`guardrails.py`):** Clamps hour arrays to $[0..23]$ unique ascending integers, clamps reduction factors to $[0.0..1.0]$, and normalizes percentage reserves to absolute kWh.
3. **Stage 3 – Mixed-Integer Linear Program** (`optimizer.py`):** Solves the global cost-minimization dispatch schedule using the COIN-OR CBC branch-and-cut solver in **< 20 milliseconds**.

6.2 Mathematical MILP Formulation

Decision Variables (for each hour $h \in \{0, 1, \dots, 23\}$):

- $g_h \geq 0$: Grid electricity import (kWh)
- $s_h \geq 0$: Solar power consumed on campus (kWh)
- $c_h \geq 0$: Energy charged into the battery (kWh)
- $d_h \geq 0$: Energy discharged from the battery (kWh)
- $u^{\text{chg}}_h \in \{0, 1\}$: Binary charging indicator
- $u^{\text{dis}}_h \in \{0, 1\}$: Binary discharging indicator
- $E_h \geq 0$: Battery State of Charge (energy level) at the end of hour h (kWh)
- $P_{\text{peak}} \geq 0$: Auxiliary variable tracking peak grid import across 24 hours

Objective Function:

- $\min_{g, s, c, d, E, u} \sum_{h=0}^{23} \left(g_h \times \text{tariff}_h \right) + 10^{-4} \times P_{\text{peak}}$
- **Primary Term:** Minimizes total grid purchase cost in Bangladeshi Taka (BDT)
 - **Regularization Term** ($10^{-4} \times P_{\text{peak}}$):** Breaks mathematical degeneracy during flat-tariff hours, smoothing the demand profile and preventing artificial spikes.

System Constraints:

1. **Hourly Campus Power Balance:**
 $g_h + s_h + d_h = \text{demand}_h + c_h \quad \forall h \in \{0, \dots, 23\}$
2. **Solar Resource Limits:**
 $s_h \leq \text{solar}_h \quad \forall h$
(If a `solar_reduction` directive applies with factor α , $s_h \leq \alpha \times \text{solar}_h$).
3. **Battery Energy Transition:**
 $E_0 = E_{\text{initial}} + c_0 - d_0$
 $E_h = E_{h-1} + c_h - d_h \quad \forall h \in \{1, \dots, 23\}$
4. **Battery Energy Bounds (Safety Reserves):**
 $E_{\text{min}} \leq E_h \leq E_{\text{max}} \quad \forall h$
($50 \text{ kWh} \leq E_h \leq 500 \text{ kWh}$)
5. **Inverter C-Rate Limits:**
 $c_h \leq C_{\text{max}} \times u^{\text{chg}}_h \quad (C_{\text{max}} = 100 \text{ kW})$

$$d_h \leq D_{\text{max}} \times u^{\text{dis}}_h \quad (D_{\text{max}} = 100 \text{ kWh})$$

6. **Mutual Exclusivity of Inverter States:**

$$u^{\text{chg}}_h + u^{\text{dis}}_h \leq 1 \quad \forall h$$
 (Guarantees that a physical battery never attempts to charge and discharge simultaneously).

7. **End-of-Day Battery Neutrality:**

$$E_{23} = E_{\text{initial}} = 250 \text{ kWh}$$
 (Prevents the optimizer from "cheating" by completely draining the battery to zero to show artificially low cost, leaving the campus defenseless for the next day).

8. **Peak Tracking Constraint:**

$$g_h \leq P_{\text{peak}} \quad \forall h$$

7. The Closed-Loop Microgrid Protection Mechanism

A frequent critique of energy optimizers is that they operate as open-loop calculators without interacting with the physical consumers. **EnergyWise** implements a closed loop:

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